

# Automaton Model of Dynamic Organization in Microtubules

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Microtubules (MT) participate in a wide range of organized dynamical activities that apparently involve information processing. Cellular automata are dynamical systems that can generate and process patterns and information by way of rules based on lattice neighbor interactions.<sup>1-3</sup> Lattice structure and apparent information processing capabilities have suggested an MT-automaton behavior based on coherent dipole oscillations within MT subunits.<sup>4-6</sup>

Various models of dynamic activities in biomolecules predict functional organization and regulation. Among these, Frohlich has theorized that nonlinear dipole excitations among subunits of biomolecular arrays result in coherent oscillations ( $10^{-10}$  to  $10^{-11}$  sec fluctuations), lower frequency metastable states, long-range cooperativity and order, and coupling of protein conformational states to dipole oscillations within protein hydrophobic regions.<sup>7,8</sup> Individual subunit conformation within a lattice array would thus depend on various factors (GTP,  $\text{Ca}^{++}$ , primary structure, lattice neighbor electrostatic interactions) influencing oscillation phase. Based on MT geometry, we have calculated MT-lattice neighbor electrostatic influences on dipole oscillations as rules for an MT-automaton computer simulation (FIGURE 1). Dimer states at each generation can be determined by neighbor states at the previous generation.

$$\text{state} = \alpha, \text{ if } \sum_{i=1}^n f(y) > 0, \text{ state} = \beta, \text{ if } \sum_{i=1}^n f(y) < 0,$$

where  $n = 7$  as the number of neighbors, and  $f(y)$  the force from the "i"th neighbor in the  $y$  direction.

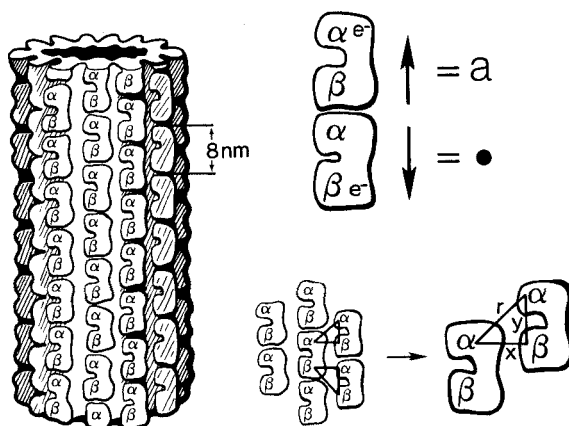
$$f(y) \propto \frac{\sin \theta}{r^2}, \sin \theta = \frac{y}{r}, f(y) \propto \frac{y}{r^3};$$

thus states may be determined by  $\sum_{i=1}^n y/r^3$  and relative values calculated from known MT structure.

FIGURE 2 shows one sequence from an "opened" MT automaton simulation in which 13 MT protofilaments are arranged horizontally. Each generation represents one Frohlich oscillation period ( $\sim$  one nanosecond). "a" represents an alpha-tubulin dipole oscillation phase, whereas dots connected by lines represent beta-tubulin oscillation phases. Generations 22, 28, 34, and 40 from a particular sequence

demonstrate stable patterns (during even-numbered generations) and a traveling kink pattern ( $\sim 80$  meters/sec), which leaves an altered "wake" (? memory). These traveling patterns are known as gliders in automaton nomenclature and have been shown by von Neumann to be capable of universal computing.

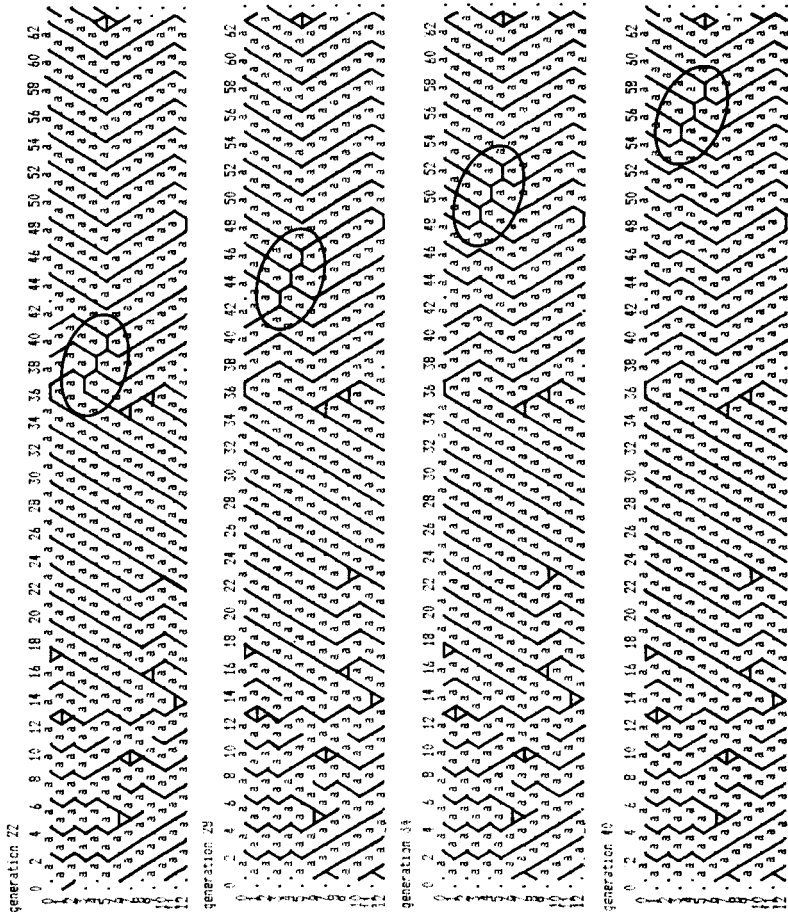
Stable and moving patterns of an MT-subunit oscillation phase could function as specific sites of ion or protein binding and transport, orchestration of biomolecular activities, as well as stored information or memory. Interaction and interference of these patterns could process information as in computers and provide "real time" biological information processing. Although the MT-automaton rules and patterns described here are somewhat arbitrary, they demonstrate that MT-automaton activities are feasible and are of potential importance in dynamic organization.



**FIGURE 1.** Left: MT structure from X-ray diffraction crystallography. Right top: MT-tubulin dimer subunits comprised of  $\alpha$  and  $\beta$  monomers. Electron occupancy of either monomer may correlate with coherent dipole oscillations and are represented as "a" for  $\alpha$ -monomer electron occupancy and dots connected by lines for  $\beta$ -monomer electron occupancy. Center bottom and right: hexagonal relations lead to neighbor rules based on lattice distances and electrostatic interactions. For neighbors shown (lower right),  $x = 5$  nm,  $y = 4$  nm,  $r = 6.4$  nm.

## REFERENCES

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**FIGURE 2.** Generations 22, 28, 34, and 40 from an MT-automaton simulation. Stable patterns have evolved during even-numbered generations. A traveling “glider” has left an altered “wake,” which may be likened to a computer operation.

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